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Case-Based Reasoning System Development
Disaster Response Decision Support System (DSS)

1. Description
   - The system stores past cases on a JSON file called disaster_cases.json. Each case includes the disaster type, flood risk, area type, estimated population, and solution which includes the evacuation method, resources to be prepared, and the capacity of available shelter.
   - The user can choose what kind of case is being reported which is run through a similarity test, comparing the current report to recorded past cases.
   - The resulting recommended actions will be recorded to the JSON file, adding to the recorded cases to be used in future disasters.
2. Explanation of similarity method
   - The system uses weighted similarity scoring to identify the most relevant past disaster case. Each case receives a score out of 100% based on four attributes: disaster type (35 points), flood risk (30 points), area type (20 points), and population similarity (15 points).
   - Matching disaster types, flood risks, and area types receive their assigned points. Population similarity is calculated by dividing the smaller population by the larger population and multiplying the result by 15. The system ranks all past cases and retrieves the case with the highest similarity score.
3. Adaptation Logic
   - After retrieving the most similar case, the system modifies its response plan to fit the new disaster scenario. It adjusts the evacuation strategy according to the disaster type, flood risk, area type, and affected population. Resource allocation is also adjusted based on population size and area characteristics. The target shelter capacity is estimated at 20% of the affected population as a simplified planning assumption. Finally, the adapted solution is saved alongside the new case in the JSON file, allowing the system to use it in future disaster assessments.
4. Reflection
   - The challenges I encountered are making the similarity scoring by deciding what aspect of the disaster should be weighed heavily compared to others and how the actions should become depending on the options entered by the user.
   - I learned that CBR is a great support decision strategy for a lot of situations that can occur repeatedly and for adapting new methods or solutions depending on past cases. For me, I think that this could be beneficial in most situations like treating injuries, diagnosing patients, and so on.

Program Output

```
--------------------------------------------------
NEW PROBLEM DESCRIPTION
--------------------------------------------------
Disaster type: Volcanic Eruption
Flood risk: Low
Area type: Rural
Affected population: 3000

SIMILARITY RANKINGS
1. Earthquake Emergency - Sample - 33.75%
2. Rural Flood Emergency - Sample - 29.0%
3. Urban Flooding - Sample - 5.62%
4. Typhoon Ursula (2019) - Sample - 4.5%
5. Typhoon Odette (2021) - Sample - 3.0%
6. New Landslide Case - 1.5%

--------------------------------------------------
RETRIEVED PAST CASE
--------------------------------------------------
Case: Earthquake Emergency - Sample
Similarity score: 33.75 %
Original evacuation: Move residents away from unsafe buildings
Original resources: Prepare first aid and rescue teams

--------------------------------------------------
ADAPTED FINAL SOLUTION
--------------------------------------------------
Evacuation plan: Follow official evacuation orders and move residents away from volcanic hazard zones. Prior
itize the most vulnerable residents
Resource allocation: Prepare essential food, water, first aid, and local emergency personnel; identify remot
e communities and available transport
Target shelter capacity: 600 people

NEW CASE SAVED SUCCESSFULLY!
File: disaster_cases.json
```

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--------------------------------------------------
UPDATED CASE BASE
--------------------------------------------------
Total cases: 7
Case 1: Typhoon Ursula (2019) - Sample | Type: Typhoon | Risk: High | Area: Coastal | Population: 10000
Case 2: Typhoon Odette (2021) - Sample | Type: Typhoon | Risk: Medium | Area: Coastal | Population: 15000
Case 3: Urban Flooding - Sample | Type: Flood | Risk: High | Area: Urban | Population: 8000
Case 4: Earthquake Emergency - Sample | Type: Earthquake | Risk: Low | Area: Urban | Population: 12000
Case 5: Rural Flood Emergency - Sample | Type: Flood | Risk: Medium | Area: Rural | Population: 5000
Case 6: New Landslide Case | Type: Landslide | Risk: High | Area: Mountainous | Population: 30000
Case 7: New Volcanic Eruption Case | Type: Volcanic Eruption | Risk: Low | Area: Rural | Population: 3000

CBR PROCESS COMPLETED.
```